The Ethical Hacker's Command Handbook

Chapter 2 – Part 9: Advanced Nmap Features (Defensive Awareness & Network Analysis)

Understand advanced Nmap features for authorized testing, troubleshooting, and network analysis.

Learning Objectives

Understand advanced Nmap options, when to use them, their limitations, and how they support troubleshooting and analysis in authorized environments.

nmap -n

Purpose: Disable DNS

What it does: Skips reverse DNS lookups.

When to use: Faster scans and lab environments.

Avoid when: Hostnames are needed.

Memory Tip: n = No DNS

nmap -R

Purpose: Force DNS

What it does: Always resolve hostnames.

When to use: Asset inventory and documentation.

Avoid when: Scan speed is more important.

Memory Tip: R = Resolve

nmap -6

Purpose: IPv6 Scan

What it does: Scans IPv6 targets.

When to use: IPv6-enabled networks.

Avoid when: IPv4-only targets.

Memory Tip: 6 = IPv6

nmap --packet-trace

Purpose: Packet Trace

What it does: Displays packets sent and received.

When to use: Troubleshooting and learning.

Avoid when: Routine scans.

Memory Tip: Trace = Watch packets

nmap -e eth0

Purpose: Select Interface

What it does: Uses a chosen network interface.

When to use: Systems with multiple interfaces.

Avoid when: Only one interface exists.

Memory Tip: e = Interface

nmap --source-port 53

Purpose: Source Port

What it does: Uses a specified source port.

When to use: Testing firewall policies in labs.

Avoid when: Default behavior is sufficient.

Memory Tip: Specify packet source port

nmap -f

Purpose: Fragment Packets

What it does: Sends fragmented probe packets.

When to use: Studying fragmentation in labs.

Avoid when: General scanning.

Memory Tip: f = Fragment

nmap --mtu 24

Purpose: Custom MTU

What it does: Sets fragment size.

When to use: Controlled networking experiments.

Avoid when: No fragmentation required.

Memory Tip: MTU = Maximum Transmission Unit

nmap --spoof-mac 00:11:22:33:44:55

Purpose: Specify MAC

What it does: Uses a chosen MAC address on local networks.

When to use: Testing local network behavior.

Avoid when: Internet scanning.

Memory Tip: MAC = Layer 2 identity

Comparison Table

Feature	Command

Disable DNS	-n
Force DNS	-R
IPv6	-6
Packet Trace	--packet-trace
Interface	-e
Source Port	--source-port
Fragmentation	-f
Custom MTU	--mtu
MAC Address	--spoof-mac

Practice Questions

1. What is the difference between -n and -R?
2. When is --packet-trace useful?
3. Why choose an interface with -e?
4. When should you use -6?
5. Why do MAC address options matter only on local networks?